Nirma University

Institute of Technology

Department of Computer Science & Engineering

Explainable AI

Lab 2: Explainability in Logistic Regression and SVM

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Contents

1 Objective **2**

2 Datasets **2**
2.1 Synthetic Customer Churn Dataset 2
2.2 Iris Dataset (Binary Subset) 2

3 Methodology **2**
3.1 Part 1: Logistic Regression 2
3.1.1 Model Description 2
3.1.2 Implementation Steps 3
3.1.3 Explainability Analysis 3
3.2 Part 2: Support Vector Machine 4
3.2.1 Model Description 4
3.2.2 Implementation Steps 4
3.2.3 Explainability Analysis 5

4 Conclusion **6**

1 Objective

The objective of this lab is to demonstrate explainability in two widely-used classification models:

1. **Logistic Regression** — interpreting coefficients as log-odds contributions
2. **Support Vector Machine (SVM)** — visualizing the decision boundary, margin, and support vectors

Both models are **intrinsically interpretable** when used with appropriate kernels and visualizations.

2 Datasets

2.1 Synthetic Customer Churn Dataset

- **Task:** Binary classification (churn vs. no churn)
- **Samples:** 1,000 (generated using `make_classification`)
- **Features (5):** `feature_1` through `feature_5`, all informative with no redundant features
- **Target:** `churn` (0 = no churn, 1 = churn)
- **Generation Parameters:** `n_informative=5`, `n_redundant=0`, `random_state=42`

2.2 Iris Dataset (Binary Subset)

- **Task:** Binary classification (Setosa vs. Versicolor)
- **Source:** `sklearn.datasets.load_iris`, filtered to classes 0 and 1
- **Samples:** 100 (50 per class)
- **Features Used (2):** Sepal length and petal length (selected for 2D visualization)
- **Target:** Species (0 = Setosa, 1 = Versicolor)

3 Methodology

3.1 Part 1: Logistic Regression

3.1.1 Model Description

Logistic Regression models the probability of the positive class using the logistic (sigmoid) function:

$$P(y = 1 \mid \mathbf{x}) = \sigma(\beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p) = \frac{1}{1 + e^{-(\beta_0 + \boldsymbol{\beta}^T \mathbf{x})}} \tag{1}$$

The coefficients β_j represent the change in **log-odds** for a one-unit increase in feature x_j:

$$\log\left(\frac{P(y=1)}{P(y=0)}\right) = \beta_0 + \sum_{j=1}^{p} \beta_j x_j \tag{2}$$

3.1.2 Implementation Steps

1. Generated synthetic binary classification data with 5 informative features
2. Trained `LogisticRegression` using the `liblinear` solver
3. Extracted and printed model coefficients and intercept
4. Plotted absolute coefficient magnitudes as a bar chart to visualize feature importance

3.1.3 Explainability Analysis

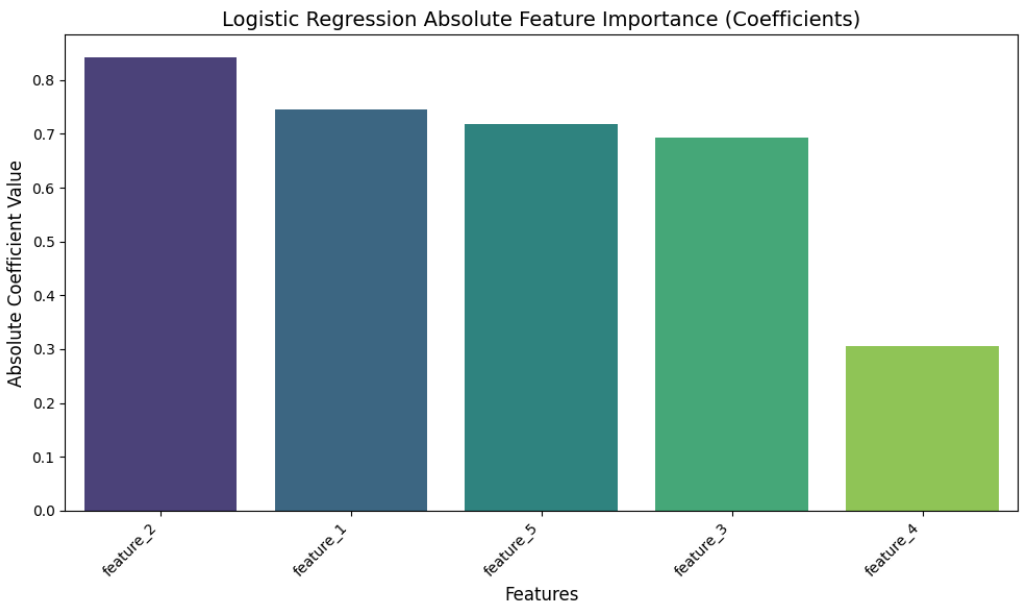

Figure 1: Logistic Regression coefficient magnitudes for the customer churn prediction model. Feature 2 has the strongest influence on churn prediction.

The trained model yielded the following coefficients:

Feature	Coefficient	Effect on Churn
feature_2	+0.8422	Increases churn probability
feature_1	−0.7461	Decreases churn probability
feature_5	−0.7187	Decreases churn probability
feature_3	+0.6936	Increases churn probability
feature_4	−0.3051	Decreases churn probability
Intercept	+0.7327	Baseline log-odds

Interpretation: A positive coefficient means that increasing the feature value increases the log-odds of churning. For example, a one-unit increase in `feature_2` increases the log-odds of churn by 0.8422, which translates to an odds ratio of $e^{0.8422} \approx 2.32$ — meaning the odds of churn more than double.

The positive intercept (0.7327) indicates that even without considering any features, there is an inherent bias toward churn in this dataset.

3.2 Part 2: Support Vector Machine

3.2.1 Model Description

A linear SVM finds the hyperplane that maximizes the margin between two classes:

$$f(\mathbf{x}) = \mathbf{w}^T\mathbf{x} + b \tag{3}$$

where $\mathbf{w}$ is the weight vector (normal to the hyperplane) and b is the bias term. The decision boundary is defined by $f(\mathbf{x}) = 0$, and the margin boundaries are $f(\mathbf{x}) = \pm 1$.

The optimization objective is:

$$\min_{\mathbf{w},b} \frac{1}{2}\|\mathbf{w}\|^2 \quad \text{subject to} \quad y_i(\mathbf{w}^T\mathbf{x}_i + b) \geq 1, \quad \forall i \tag{4}$$

Support vectors are the data points that lie exactly on the margin boundaries ($y_i f(\mathbf{x}_i) = 1$).

3.2.2 Implementation Steps

1. Loaded the Iris dataset and filtered to two classes (Setosa and Versicolor)
2. Selected two features (sepal length, petal length) for 2D visualization
3. Trained `SVC(kernel='linear')` from scikit-learn
4. Extracted the weight vector $\mathbf{w}$ and intercept b
5. Computed the decision boundary line: $x_2 = -\frac{w_1}{w_2}x_1 - \frac{b}{w_2}$
6. Computed margin lines at $\pm\frac{1}{w_2}$ offset from the decision boundary
7. Plotted data points (colored by class), support vectors (circled), decision boundary, and margins

3.2.3 Explainability Analysis

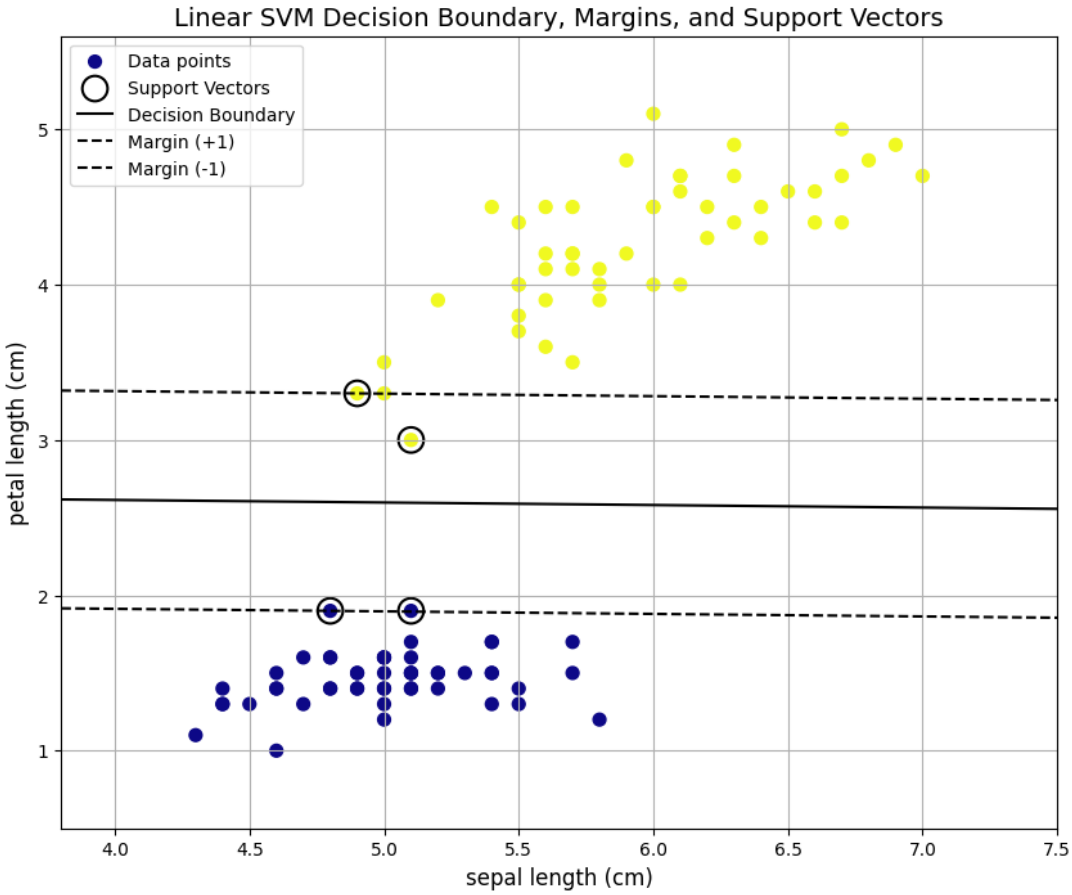

Figure 2: SVM decision boundary visualization. Data points are colored by class (Setosa vs. Versicolor). Support vectors (circled) define the margin. The dashed lines indicate the margin boundaries.

Interpretation:

- The **decision boundary** (solid line) separates Setosa from Versicolor flowers. Points above the line are classified as one class, points below as the other.
- The **support vectors** (circled points) are the critical samples that define the boundary. These are the "hardest to classify" examples — removing any support vector would shift the boundary.
- The **margin width** indicates model confidence. A wider margin means the two classes are well-separated. In this binary Iris problem, the classes are highly separable, resulting in a clear wide margin.
- The **weight vector w** reveals that petal length has a larger weight than sepal length, indicating it is the more discriminative feature. This aligns with botanical knowledge — Setosa flowers have distinctly shorter petals.

4 Conclusion

This lab demonstrated that both Logistic Regression and linear SVMs are inherently interpretable:

- **Logistic Regression** coefficients directly quantify each feature's contribution to the prediction in terms of log-odds. This makes it easy to identify the most influential factors and their direction of effect.
- **Linear SVM** provides geometric interpretability through its decision boundary, support vectors, and margin. The weight vector reveals feature importance, and the support vectors identify the most informative training examples.

Both methods are widely used in domains requiring transparent decision-making, such as healthcare and finance, where practitioners need to understand and justify model predictions.